Course Title: **SWE 333: Software Verification & Testing**

Prerequisites: Basic programming knowledge

Course Objectives:

- Understand the principles and fundamentals of software testing.
- Learn various software testing methodologies and techniques.
- Design effective test cases and manage testing activities.
- Use testing tools and apply automation for real-world scenarios.

Course Outline

Module 1: Introduction to Software Testing

1. Importance of Software Quality
2. Software Testing Principles and Objectives
3. Errors, Defects, and Failures
4. Testing Process and Life Cycle
5. Role of a Software Tester

Module 2: Testing Techniques

White-Box Testing

1. Statement Coverage
2. Decision and Condition Coverage
3. Data Flow Testing
4. Mutation Testing

Black-Box Testing

1. Equivalence Partitioning
2. Boundary Value Analysis
3. Cause-Effect Graphing

Module 3: Levels of Testing

1. Unit Testing
2. Integration Testing
3. System Testing
4. Acceptance Testing

Module 4: Test Case Design and Management

1. Test Case Design Principles
2. Writing Effective Test Cases
3. Test Planning and Estimation
4. Test Reporting

Module 5: Debugging

1. Debugging Process
2. Debugging Techniques
3. Debuggers

Module 6: Test Automation

1. Introduction to Automation
2. Automation Tools Overview (e.g., Selenium, QTP)
3. Writing Test Scripts
4. Benefits and Limitations of Automation